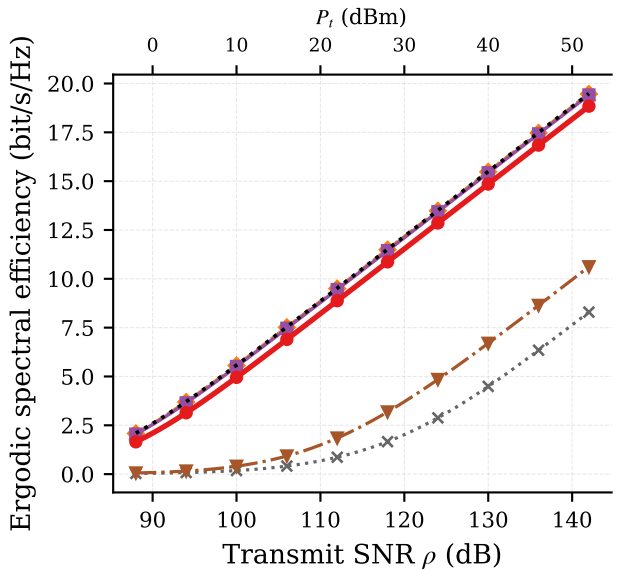


(a) Achievable rate

Legend for (a):

- $\cdots \times \cdots$ No RIS (blocked)
- $\cdots \blacktriangledown \cdots$ Alternating Opt.
- $\cdots \blacktriangleright \cdots$ Random phases
- $\cdots \blacklozenge \cdots$ SCA
- $\cdots \blacksquare \cdots$ Centralized DL
- $\cdots \bullet \cdots$ Fed-RIS (proposed)
- $\cdots \cdots$ Perfect-CSI MRC (bound)

(b) Gain over the blocked direct link